Stable Hulls as Imprint Integration: An Interpretive Bridge from Filtered Traces to Self-Reading World-Models

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Abstract

This paper develops a programme-level bridge between the imprint vocabulary of *The World as Imprint* and the accessible-world machinery distributed across Foundation II, Foundation III, and *The Brain as a Plastic Observable-Algebra Network*. Its claim is conditional and architectural. The decoder-relative imprint may be small or null, so neither carrying nor stable-hull formation is guaranteed. When readable traces are present, the paper proposes stable hulls as integration regimes through which selected traces may become accessible, retainable, reportable, and eventually available for organized self-reading. It does not identify stable hulls with consciousness or provide sufficient conditions for first-person subjectivity.

The central thesis is likewise conditional: a finite observer does not encounter a full candidate imprint-field, but only those readable traces, if any, that enter an accessible algebra, stabilize under repeated uptake, and become integrated into a self-world map. Stable hulls are proposed as a middle regime between candidate imprint-bearing structure and uneven self-reading. They describe how readable traces may become a usable world before becoming a reflexively read self. The contribution is internal architectural clarification for an existing multi-paper programme, not a new result in cognitive science or neuroscience.

Keywords: stable hull, imprint integration, accessible algebra, self-model, memory, self-reading, world-model

1 Introduction: From Imprint-Field to Accessible World

The World as Imprint argued that the emergent world can be read as an imprint-field: a domain of structures insofar as they carry readable marks of the generative constraints that shape them [1]. That heuristic explicitly allowed the decoder-relative imprint to be small or null; carrying is not guaranteed. But the same paper also insisted that carrying is not the same as reflexive recognition. Not every structure that bears a readable mark thereby counts as a mind. What remained underdeveloped was the middle regime between

these two claims. How do readable traces become an accessible world for a finite observer before they become the object of organized self-reading?

This paper develops that middle regime through a distinct cognitive object, the integration workspace $\mathcal{W}_{\text{int}}$. It is not a mysterious substance or a synonym for consciousness. It is a programme-level integration regime in which filtered traces are proposed to become jointly available for memory, report, coordinated uptake, and self-location.

The bridge matters because without it the path from imprint-carrying to self-reading remains too abrupt. One would then have only two levels: candidate imprint-bearing structure on one side and mind on the other. But the actual architecture of finite observers is richer. Before a system can read itself as imprint-bearing, it must inhabit a world that is stable enough to support retained salience, cross-temporal coordination, and self-world differentiation. That world is not the whole imprint-field. It is a stabilized accessible sector carved out of it.

Earlier papers use similar hull language for different typed objects. Foundation I has a source-algebraic intersection object, Foundation II has an external shared accessible sector, and the Brain Anchor discusses an internal cognitive workspace [2, 3, 5]. The present paper concerns only the third type, denoted $\mathcal{W}_{\text{int}}$. It does not identify or derive it from either Foundation object.

> filtered imprints $\rightarrow$ accessible organization $\rightarrow \mathcal{W}_{\text{int}} \rightarrow$ retained memory $\rightarrow$ self-world model.

Several guardrails are necessary at the outset. First, $\mathcal{W}_{\text{int}}$ is proposed as a precondition for organized self-reading, not a sufficient condition for consciousness. Second, memory integration is not personal survival or identity transfer. Third, this paper does not determine which animals or machines are conscious. Fourth, the self-reading threshold is treated elsewhere [1]. The present paper is an interpretive companion, not an independent contribution to self-model theory or neuroscience.

The paper proceeds accordingly. Section 2 separates the three hull namespaces. Section 3 introduces filtered imprints and accessible organization. Section 4 interprets $\mathcal{W}_{\text{int}}$ as a cognitive integration regime. Later sections treat memory, self-world modeling, interfaces to other papers, and limitations.

2 Hull Namespace and Type Registry

The phrase "stable hull" has been used for three objects with different domains and types:

$\mathcal{H}_{\text{st}}$ the Foundation I source-algebraic stable hull, defined by an intersection construction in the source/semigroup setting [2];

$\mathcal{H}_{12}$ the Foundation II external shared accessible sector for two independently specified source chains [3];

$\mathcal{W}_{\text{int}}$ the internal cognitive integration workspace used in this bridge and associated with the Brain Anchor programme [5].

No equality, inclusion, or canonical map among these three objects is established. In particular, $\mathcal{H}_{12}$ is not an internal brain workspace, and $\mathcal{W}_{\text{int}}$ is not the Foundation I intersection $\mathcal{H}_{\text{st}}$.

Definition 2.1 (Integration workspace)**.** *The* integration workspace $\mathcal{W}_{\text{int}}$ *is a programme-level cognitive regime in which a family of filtered traces is sufficiently persistent, mutually coherent, and recurrently available to support world-level uptake, retention, and coordinated redescription.*

Three phrases in the definition matter.

First, "cognitive regime" means that $\mathcal{W}_{\text{int}}$ is typed relative to internal uptake by a finite system, unlike the external agreement object $\mathcal{H}_{12}$. Second, "mutually coherent" means that retained traces support a structured world-picture. Third, "recurrently available" means that traces can be compared, updated, and reused.

This definition is an architectural proposal, not a consequence of Foundation II. The imprint heuristic also does not guarantee that traces exist everywhere: relative to a source family and decoder class the imprint value may be zero. The question here is conditional—when readable traces exist, which become organized enough to form a usable world-model?

3 Filtered Imprints and Accessible Algebra

The first bridge paper argued that coarse-graining can be read as imprint filtering: not every lower-level feature survives scale change, and those that do survive constitute scale-stable residues. But even after such filtering, the resulting imprint-field is still too large to be inhabited in full by any finite observer. A second selection is therefore necessary. This second selection is not primarily a matter of scale; it is a matter of access.

Definition 3.1 (Accessible imprint)**.** *An* accessible imprint *is a filtered trace that lies within the observer-relative regime of uptake, retention, comparison, or action-guidance.*

This definition is deliberately observer-relative without becoming subjectivist. The point is not that accessibility is arbitrary or purely private. The point is that finite systems encounter only a structured subset of all available marks. An accessible algebra is one way to formalize this subset in the broader programme [2, 3]. For present purposes, the essential idea is simpler: the world of a finite observer is formed by those traces that can enter stable uptake.

This gives a two-step selection picture:

1. coarse-graining selects which lower-level features survive as scale-stable imprints;
2. accessibility selects which of those imprints can enter an observer's usable world.

The proposed workspace $\mathcal{W}_{\text{int}}$ lives at the second step. It does not explain why a trace survives scale change. It names the further cognitive integration problem for a readable trace.

This is also where the bridge differs from standard cognitive-theory usage. It does not claim to replace global workspace, predictive processing, or enactive models. It claims only to identify an architectural slot that these literatures often presuppose in different ways: the transition from mere accessible traces to a stable, reusable world-organization.

4 The Integration Workspace $\mathcal{W}_{\text{int}}$

It is tempting to think that once traces are accessible, a world is already present. That is too quick. Accessibility alone may still be fragmented, transient, or mutually inconsistent. What is needed for worldhood is integration.

4.1 Why accessibility is not enough

A finite system can register signals without organizing them into a stable world. It may encounter accessible imprints but fail to build an integration workspace $\mathcal{W}_{\text{int}}$. The distinction marks the difference between signal-response and the proposed architecture of inhabitable worldhood.

4.2 Integration conditions

Without pretending to state a theorem, the following conditions capture what the $\mathcal{W}_{\text{int}}$ bridge is trying to mark:

Condition	**Role in $\mathcal{W}_{\text{int}}$ integration**
persistence	the same or comparable traces remain available across time
coherence	retained traces support a jointly navigable picture rather than disconnected shards
selective retention	some traces are stabilized while others are dropped or backgrounded
cross-context usability	traces can guide report, action, or re-identification across situations
self-world differentiation	the regime can support a distinction between world-features and self-locating features

None of these conditions is by itself equivalent to consciousness. Together they specify something weaker but still crucial: a world-forming regime in which accessible traces become jointly organized. These conditions also resonate with global-availability accounts of conscious access [6, 7]. In that sense, the table is best read as an interface map rather than as a rival empirical theory.

4.3 Integration workspace and reportable worldhood

The integration workspace is proposed as a precondition for reportable worldhood. A system may have filtered inputs without such a workspace; in that case the bridge supplies no stable map for comparison or re-entry. Even when $\mathcal{W}_{\text{int}}$ is present, reflexive self-reading does not follow.

5 Memory as Retained Imprint Integration

Memory illustrates why a distinct cognitive integration interface is useful. It is selective retention for later integration, not an instance of the external agreement sector $\mathcal{H}_{12}$.

5.1 Retention is selective

No finite system copies a full source description. Memory operates by selective retention: some traces stabilize, others decay or are overwritten, and some remain only as biasing conditions. What survives may be re-entered into $\mathcal{W}_{\text{int}}$.

5.2 Memory is integrative, not archival

The metaphor of archive is misleading. Memory is not a warehouse of preserved states. It is better understood as retained imprint integration. A memory trace matters not because it exists in isolation, but because it can be recruited back into a larger world-model. It contributes to expectations, error-correction, self-location, and salience structure. Within predictive-processing accounts, retained traces function as priors shaping anticipation and prediction-error correction [10].

The Brain Anchor paper motivates an internal workspace interpretation through plasticity and recurrent world-modeling [5]. The present paper does not verify those claims or import them as a theorem; their consistency with the namespaced $\mathcal{W}_{\text{int}}$ is a separate interface question. Plastic stabilization is used here only as a candidate illustration of retained world-structure.

5.3 No claim about identity transfer

The memory discussion also demands a guardrail already familiar from *The World as Imprint*. Retention of traces does not establish persistence of the same first-person subject elsewhere [1]. The fact that effects propagate, habits persist, or structures re-enter later organization should not be confused with personal identity transfer. In the present framework, memory integration concerns world-organization, not metaphysical survival.

6 The Self-World Model as Organized Imprint Map

Once an integration workspace $\mathcal{W}_{\text{int}}$ is stipulated, a further transition becomes possible: construction of a self-world model. This is an architectural ordering, not a theorem of cognitive necessity.

6.1 Map, not inner substance

The phrase "self-world model" risks misunderstanding if treated too quickly. It does not refer to a ghostly entity observing the world from behind the scenes. It names an organized map in which world-features, bodily regularities, action possibilities, and self-locating coordinates become mutually articulated. In imprint language, it is a way of organizing integrated traces into a differentiated picture. This formulation aligns with the self-model theory of subjectivity [8], while still treating the present paper's claim as an architectural redescription rather than a reconstruction of phenomenal transparency.

6.2 Why the self-model comes after the workspace in this bridge

Within this proposed bridge, a self-model is placed after a sufficiently stable world-map. Accordingly $\mathcal{W}_{\text{int}}$ is a stipulated architectural precondition for the self-model, not a result

inherited from $\mathcal{H}_{\text{st}}$ or $\mathcal{H}_{12}$. The bridge stops before the self-reading threshold.

6.3 Organized imprint map

The self-world model can therefore be described in a sentence:

> A self-world model is an organized imprint map in which retained traces are structured into a navigable distinction between self-locating, world-locating, and action-guiding organization.

This formulation does not yet say that anything "feels like something." It says only that a finite regime has reached the point at which world-organization and self-location are jointly stabilized. Whether the integration is best described in narrative terms or in transparent representational terms remains open [9, 8].

7 Relation to Existing Papers

The present bridge is useful only if its place in the larger body of work is explicit.

7.1 Relation to Foundation II

Foundation II studies the external shared accessible sector $\mathcal{H}_{12}$ across independently specified source-side chains [3]. That object is not the internal cognitive workspace $\mathcal{W}_{\text{int}}$. The former proposal to import Foundation II as a formal derivation or de-tautologization of the present workspace is retracted. At most, external agreement motivates an analogy; no typed bridge has been constructed.

7.2 Relation to Foundation III

Foundation III adds conditional feedback structure at the source-manifest interface [4]. The present paper imports no feedback theorem. It uses historically shaped organization only as a programme-level analogy for retention and re-entry in $\mathcal{W}_{\text{int}}$.

7.3 Relation to Brain Anchor

The Brain Anchor paper offers a candidate biological realization of an internal integration workspace [5]. The present paper uses the shared symbol $\mathcal{W}_{\text{int}}$ for that cognitive type but does not verify the published paper's internal consistency or identify it with either Foundation hull. That verification is a separate Gate 2 interface task.

7.4 Relation to *The World as Imprint*

Finally, the relation to *The World as Imprint* is direct. That paper distinguished decoder-relative carrying, which may be small or null, from uneven recognition [1]. The present paper fills the middle gap. It explains how carrying becomes accessible worldhood before it becomes reflexive recognition.

8 Output to the Self-Reading Threshold

The bridge can now be stated in its cleanest form:

> Within this bridge, $\mathcal{W}_{\text{int}}$ is a proposed precondition for organized self-reading, not a sufficient condition for consciousness.

This sentence is the paper's most important guardrail. It says what the paper contributes and where it must stop.

What the paper contributes is a clearer picture of the middle architecture:

1. filtered imprints survive scale change;
2. some of them enter accessible uptake;
3. some of those are proposed to enter $\mathcal{W}_{\text{int}}$;
4. that workspace supports the architectural description of retained memory and a self-world model;
5. only then does the question of self-reading threshold arise.

What the paper does not contribute is a theory of why first-person experience accompanies such organization. Nor does it provide a criterion by which all borderline biological or artificial cases can be settled. Those are later or different questions.

This is why the paper's stopping point matters. It stops at the integration of filtered imprints into an accessible self-world model; the self-reading threshold itself is treated elsewhere [1]. That is not a deficiency. It is the correct division of labor.

9 Limitations

Several limitations should remain explicit.

First, the paper is programme-level. It does not construct $\mathcal{W}_{\text{int}}$ from arbitrary imprint-fields.

Second, the terminology of hull, access, retention, and model remains partly architectural rather than fully operationalized. Further formal work would need metrics for integration depth, accessibility bandwidth, and retention stability.

Third, the bridge remains neutral on substrate. It does not assume that only brains can realize $\mathcal{W}_{\text{int}}$, but it certifies no biological or non-biological system as conscious.

Fourth, the paper does not decide whether systems realizing the proposed workspace cross the self-reading threshold. The two questions remain separate.

Fifth, the paper should not be read as offering a new empirical discriminator among existing self-model theories. Its role is interpretive and organizational: it clarifies how filtered imprints, accessible organization, and self-world mapping fit together inside the larger programme.

Despite those limitations, the namespace makes the proposed transition more precise without claiming formal inheritance:

> readable imprints $\rightarrow$ accessible world $\rightarrow$ $\mathcal{W}_{\text{int}}$ $\rightarrow$ self-world model $\rightarrow$ self-reading threshold.

This remains an interface architecture, not a theorem that the workspace exists or produces a mind.

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